

School of Engineering
Department of Computer and Communication Engineering
Semester: Fall 2012 – 2013
Course: CENG415 – Communication Networks

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Final Examination

Date: Monday February 11, 2013

Time: 9:00 – 11:00 (120 minutes)

Student Name: _____ **Student ID:** _____ **Section:** _____

There are **4 questions** in the booklet each has several parts, please answer all parts of the **4** questions to the best of your ability.

Marking Scheme:

Questions	Weight	Mark	Covered PC	Coverage %
Question 1	35 points		PC-2, PC-3, PC-5	40, 30, 30
Question 2	20 points		PC-1, PC-2, PC-3	50, 20, 30
Question 3	25 points		PC-3, PC-4, PC-5,	20, 60, 20
Question 4	20 points		PC-3, PC-4	60, 40
Total	100 points			

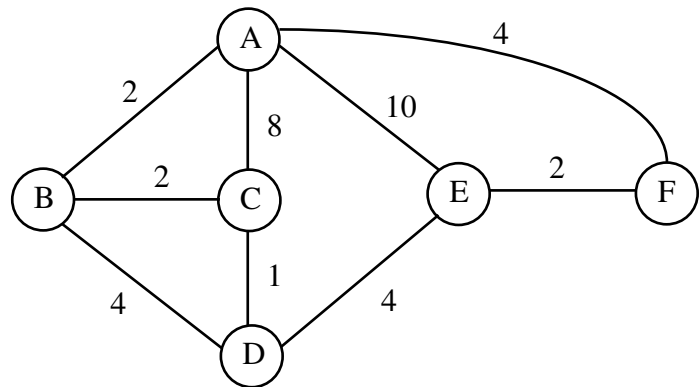
1. This booklet contains **12** pages including this one. Make sure all these pages are attached.
2. Closed book examination.
3. Do not take the staple out. The exam booklet must remain intact.
4. Cheating penalty will be an “F” grade on the exam.
5. Programmable calculators are not allowed.
6. Mobile phones/devices are to be turned off and stowed away.

Good luck

Question 1: Routing, fragmentation and NATing (35 points)

Part A: Shortest path (11 points)

1. Consider the following network:
The numbers on links between the nodes represent the costs corresponding to using these links.



- a. Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from A to all other nodes by completing the distance and predecessor tables below. (6 points)

{ 3 points for each table }

Distance Table					Predecessor Table				
B	C	D	E	F	B	C	D	E	F
2	8	∞	10	4	A	A	-	A	A
	4	6	10	4		B	B	A	A
		5	10	4			C	A	A
		5	6					F	
			6					F	

- b. From the results above, show the shortest path between node A and all the other nodes with their respective costs: (5 points) { 1 point for each shortest path: 0.5 for the cost and 0.5 for the path }

Shortest path between A and B: Cost 2 Path A→B

Shortest path between A and C: Cost 4 Path A→B→C

Shortest path between A and D: Cost 5 Path A→B→C→D

Shortest path between A and E: Cost 6 Path A→F→E

Shortest path between A and F: Cost 4 Path A→F

Part B: Fragmentation and NATing (24 points)

1. A 1500-byte IP datagram must be fragmented into 660-byte fragments.

- a. How many fragments are generated? (1 point)

3 fragments

- b. Using the information below of the Original datagram, fill in the table below to show the values of the Length, ID, Flags and Offset in the resulting fragments (use as much rows as you need). (6 points) {0.5 point for each value in the table}

Original

Length	ID	Flags	Offset
1500	123456789	0	0

Fragments

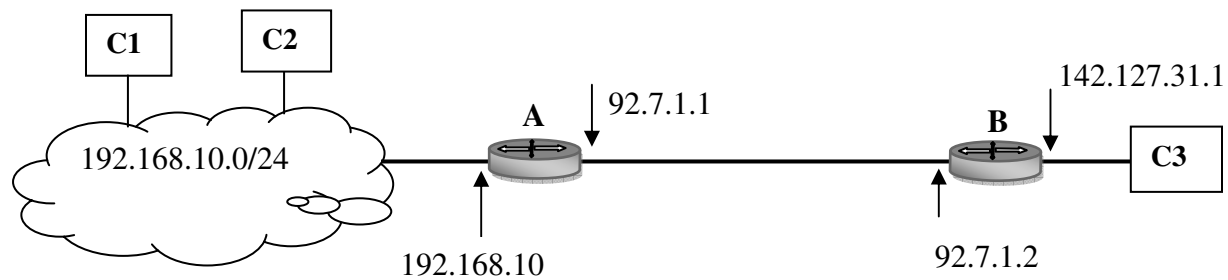
Length	ID	Flags	Offset
660	123456789	1	0
660	123456789	1	80
220	123456789	0	160

- c. Assuming that the original 1500-byte IP datagram corresponds to a response from an HTTP server, what will be at the destination, after the reassembly of the fragments, the length in bytes of the received packet at the following layers and why? (3 points)
- Network layer: 1500 bytes (same length as the original datagram) {1 point}
 - Transport layer: 1480 bytes (subtract the IP header of 20 bytes) {1 point}
 - Application layer: 1460 bytes (subtract the TCP header of 20 bytes) {1 point}

2. Consider the following network where computers C1 and C2 run a client application and C3 runs a server application with the following information. Router A is capable of Network Address Translation.

	C1	C2	C3
IP	192.168.10.1	192.168.10.2	142.127.31.14
MAC	00-1E-33-B7-B1-A2	00-1E-33-B7-B1-B2	00-1F-81-00-02-50
Port	3000	4000	80

	Router A: interface 1	Router A: interface 2	Router B: interface 1	Router B: interface 2
IP	192.168.10.5	92.7.1.1	92.7.1.2	142.127.31.1
MAC	00-50-56-C0-00-08	00-50-56-C0-00-18	00-50-56-C0-00-01	00-50-56-C0-00-BB



- a. Fill in the following information in the headers of the frame leaving C1. (3 points)

{0.5 point for each value in the table}

Destination MAC	Source MAC	Source IP	Destination IP	Source Port	Destination Port
00-50-56-c0-00-08	00-1E-33-B7-B1-A2	192.168.10.1	142.127.31.14	3000	80

- b. Fill in the following information in the headers of the frame leaving C2. (3 points)

{0.5 point for each value in the table}

Destination MAC	Source MAC	Source IP	Destination IP	Source Port	Destination Port
00-50-56-C0-00-08	00-1E-33-B7-B1-B2	192.168.10.2	142.127.31.14	4000	80

- c. Fill in the following information in the headers of the frame, arriving from C1 and leaving the Router A. (3 points)

{0.5 point for each value in the table}

Destination MAC	Source MAC	Source IP	Destination IP	Source Port	Destination Port
00-50-56-C0-00-01	00-50-56-C0-00-18	92.7.1.1	142.127.31.14	3001	80

- d. Fill in the following information in the headers of the frame, arriving from C2 and leaving the Router A. (3 points)

{0.5 point for each value in the table}

Destination MAC	Source MAC	Source IP	Destination IP	Source Port	Destination Port
00-50-56-C0-00-01	00-50-56-C0-00-18	92.7.1.1	142.127.31.14	4001	80

- e. Write the translation table in the Router A. (2 points)

{1 point for each row}

192.168.10.1: 3000 92.7.1.1:3001
 192.168.10.2: 4000 92.7.1.1:4001

Question 2: Multiple choices (20 points)

Choose the correct answer. Only one answer is correct in each question. Each multiple choices question is worth 1 point. All answers should be reported to the table at the end of this question. Only the answering table will be corrected.

1. Which of the following may be the result of using a Web proxy?
 - a. Shortened delay
 - b. Higher uplink bandwidth requirement at the server
 - c. Fresher webpage
 - d. Higher downlink bandwidth requirement at the client
2. When a TCP segment arrives at a host, in order to direct the segment to the appropriate socket, the operating system stack uses:
 - a. The destination IP address
 - b. The source port number
 - c. The source IP address
 - d. All of the above
3. Which of the following is true about HTTP, FTP and SMTP?
 - a. HTTP uses UDP as transport protocol
 - b. SMTP is a mail access protocol
 - c. HTTP sends control messages out of band
 - d. FTP sends control messages out of band
4. What functionalities do TCP and UDP have in common?
 - a. Reliable transfer
 - b. Flow control
 - c. Multiplexing
 - d. Connection establishment
5. Which of the following is true about TCP's reliable data transfer?
 - a. TCP may retransmit packets upon receiving duplicate acknowledgements
 - b. TCP may retransmit packets upon timer timeout events
 - c. TCP uses cumulative acknowledgement
 - d. All of the above
6. Flow control takes place at which layer(s)?
 - a. Network Layer
 - b. Transport Layer and Link Layer
 - c. Transport Layer only
 - d. Link Layer only
7. Which layer translates between physical addresses and logical addresses?
 - a. Network Layer
 - b. Link Layer
 - c. Application Layer
 - d. None of the above

8. Which device acts as a smart hub connecting computers into a network, and separates your network from all other networks?
- a. Router
 - b. Firewall
 - c. Switch
 - d. Hub
9. When a UDP segment arrives at a host, in order to direct the segment to the appropriate socket, the operating system uses:
- a. The destination port number
 - b. The source IP address
 - c. The source port number
 - d. The destination MAC address
10. IPv6 is different from IPv4 because
- a. It does not define fragmentation
 - b. It does not contain any optional header bytes in the datagram format
 - c. It uses 128 bits for addressing
 - d. All of the above
11. ICMP (Internet Control Message Protocol), is:
- a. an application layer protocol
 - b. a transport layer protocol
 - c. a network layer protocol
 - d. a data link layer protocol
12. The network 200.23.16.0/23 can accomodate
- a. 256 hosts
 - b. 512 hosts
 - c. 510 hosts
 - d. 8 hosts
13. With the old classful addressing method of IPv4 protocol, the IP address: 172.16.254.1 corresponds to:
- a. Class A
 - b. Class B
 - c. Class C
 - d. Class D
14. Which multiple access protocol is used in Ethernet?
- a. CSMA/CA
 - b. CSMA/CD
 - c. Slotted Aloha
 - d. TDMA
15. The subnet mask identifies whether the destination address is local or remote through:
- a. Subnetting
 - b. ANDing
 - c. Broadcasting
 - d. Demultiplexing

16. Which device connects multiple computers in a network?
- Network interface card
 - Ethernet card
 - Network hub**
 - None of the above
17. Which device acts as a smart hub connecting computers into a network, and separates your network from all other networks?
- Router**
 - Firewall
 - Switch
 - Hub
18. A message of 1500 bytes has been fragmented into 3 IP datagrams. The total size of the 3 IP datagrams together at the destination before being reassembled will be:
- 1540 bytes
 - 1580 bytes**
 - 1520 bytes
 - 1560 bytes
19. Each station on an Ethernet network has a unique address built in its network interface card (NIC) with a length of:
- 16 bits
 - 32 bits
 - 48 bits**
 - 128 bits
20. A router running distance-vector routing knows:
- nothing
 - the entire network's topology
 - the distance to destinations through its immediate neighbors**
 - the cost of all links in the network

Answering table { 1 point for each correct answer }

Question	Answer
1	a
2	d
3	d
4	c
5	d
6	b
7	b
8	a
9	a
10	d

Question	Answer
11	c
12	c
13	b
14	b
15	b
16	c
17	a
18	b
19	c
20	c

Question 3: Subnetting, routing and programming (25 points)

1. Consider the following IP addresses of three networks:

Network 1: 134.34.57.0/24

Network 2: 134.34.56.0/24

Network 3: 134.35.0.0/16

Let the IP addresses of six hosts be given as follows:

Host A: 134.35.57.19

Host B: 134.34.56.7

Host C: 134.35.0.7

Host D: 134.34.57.2

Host E: 134.34.0.250

Host F: 134.34.56.256

Determine to which network (Network 1, 2 or 3) do the different hosts A, B, C, D, E and F belong?
Are there any hosts that do not belong to any of the networks above? (6 points)

{ 1 point for each Host }

Host A belongs to Network 3.

Host B belongs to Network 2.

Host C belongs to Network 3.

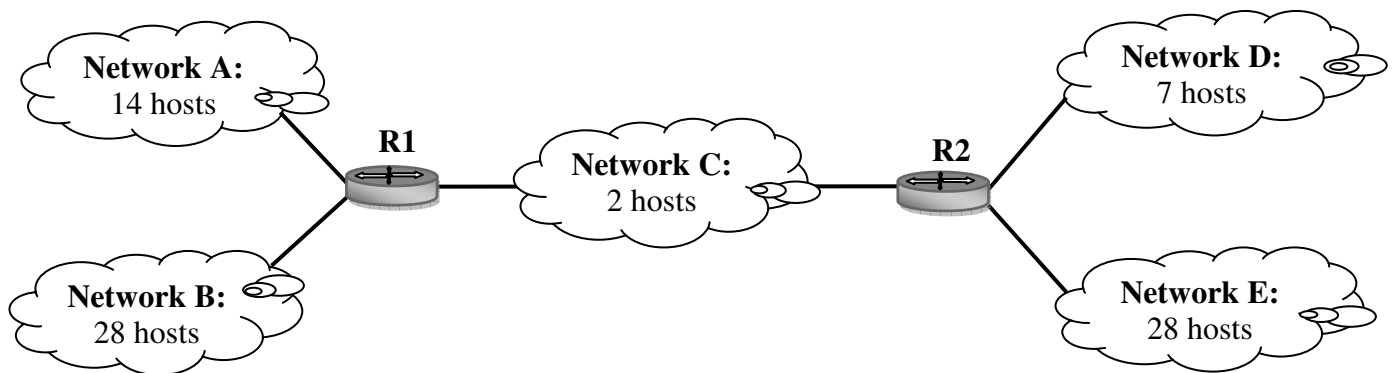
Host D belongs to Network 1.

Host E does not belong to any of the networks above.

Host F does not have a valid IP address.

2. Given the network address 204.15.5.0/24:

- a. Subnet the network in order to create the five networks with the host requirements shown in the figure below. Assign the minimum possible number of IP addresses to each of the networks A, B, C, D and E. Give the network address, the subnet mask and the address range for each of the networks. How much available IP addresses are not used? (10 points)



One optimal way to assign the subnets is:

- Network B: 204.15.5.0/27 host address range 1 to 30 {2 points: 1 for the network address and one for the range}
- Network E: 204.15.5.32/27 host address range 33 to 62 {2 points: 1 for the network address and one for the range}
- Network A: 204.15.5.64/28 host address range 65 to 78 {2 points: 1 for the network address and one for the range}
- Network D: 204.15.5.80/28 host address range 81 to 95 {2 points: 1 for the network address and one for the range}
- Network C: 204.15.5.96/30 host address range 97 to 98 {2 points: 1 for the network address and one for the range}

Remaining addresses: 204.15.5.100 to 204.15.5.255 (156 IP addresses) {Bonus: +1 for correct answer}

- b. Complete the following routing tables of Routers R1 and R2: (4 points)
{0.5 point for each value in the table}

Routing table of R1

Destination	Next hop	Number of hops
Network C	Direct	0
Network A	Direct	0
Network B	Direct	0
Network D	R2	1
Network E	R2	1

Routing table of R2

Destination	Next hop	Number of hops
Network C	Direct	0
Network A	R1	1
Network B	R1	1
Network D	Direct	0
Network E	Direct	0

- c. Complete the following commands in Packet tracer to configure the routing table of Router R1 with RIP. (4 points) {0.5 point for each row}

```

Router1> enable
Router1# configure terminal
Router1(config)# router rip
Router1(config-router)# network A IP (depending on subnetting in part a)
Router1(config-router)# network B IP (depending on subnetting in part a)
Router1(config-router)# network C IP (depending on subnetting in part a)
Router1(config-router)# exit
Router1(config)# exit

```

- d. What is the instruction needed in Packet tracer, so that the configuration remains even if the router is powered off? (1 point)

copy running-config startup-config

Question 4: Data Link Layer (20 points)

1. Consider a 10Mbps shared Ethernet segment of length 500 m with Station A at one end, Station B at the other, and Station C right in the middle. The speed of the signal over the cable is 2×10^8 m/s.

- a. Assume that Station A starts transmitting at time = 0, at what time does the first bit arrive to Station B? (1 point)

Propagation time = $T_{\text{prop}} = (\text{Distance from A to B}) / (\text{Speed}) = (500) / (2 \times 10^8) = 2.5 \mu\text{sec}$

- b. B sends a frame exactly when it receives the first bit from the frame coming from A and a collision occurs.

- i. At what time does Station B detect the collision? (1 point)

B detects collision directly at the arrival of the first bit from A $\rightarrow$ at $T_{\text{prop}} = 2.5 \mu\text{sec}$

- ii. At what time does Station A detect the collision? (1 point)

A detects the collision after one T_{prop} from the collision $\rightarrow$ at $2 \times T_{\text{prop}} = 2 \times 2.5 = 5 \mu\text{sec}$

- iii. At what time does Station C detect the collision? (1 point)

C detects the collision after $(1/2) T_{\text{prop}}$ from the collision $\rightarrow$ at $1.5 \times T_{\text{prop}} = 1.5 \times 2.5 = 3.75 \mu\text{sec}$

- c. To avoid the presence of collisions in the network, another multiple access protocol can be used such as TDMA or Polling. Give one disadvantage for each of these 2 types of MAC protocols. (2 points) {1 point for one disadvantage of each protocol}

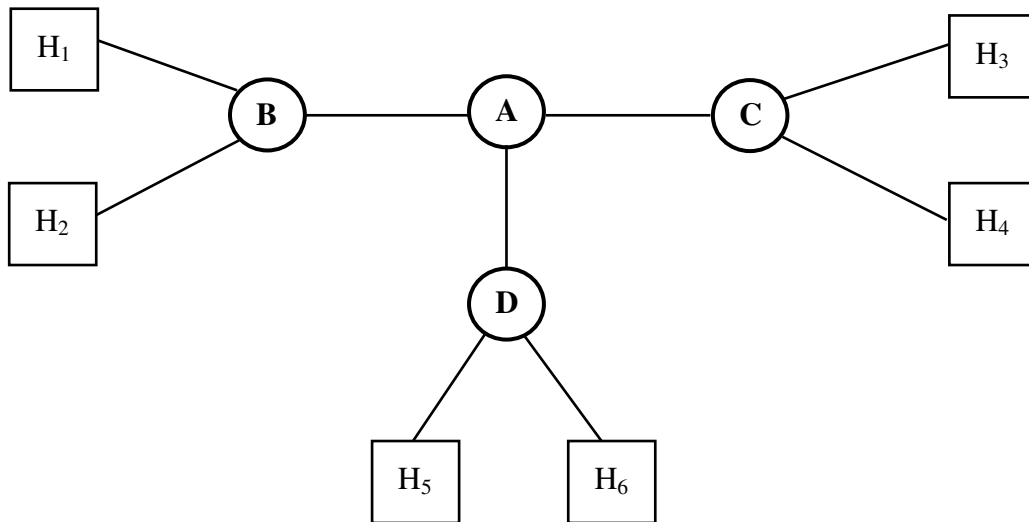
- TDMA disadvantages (1 answer among):

- A node must always wait for its turn in the transmission sequence, even when it is the only node with a frame to send
- Waste of channel capacity because unused slots are idle

- Polling disadvantages (1 answer among):

- Polling overhead
- Latency
- Single point of failure

2. Consider the network topology in the figure below, where:
- H_n ($n = 1, 2, 3, 4, 5$ and 6) represents a host machine.
 - A, B, C and D can be either hubs or switches.



- a. Suppose A, B, C and D are hubs (repeaters). Assuming that the sender does not hear the packet it sends, list the hosts that can overhear a packet sent from H_1 destined to H_4 . (3 points)
{ Subtract 0.5 point for each missed host or wrong host }
Solution: H2, H3, H4, H5, H6
- b. Assume that A is a switch (learning bridge) which knows where the hosts are (i.e. has a filled in bridging table) and B, C and D are hubs (repeaters). List the hosts that can overhear a packet sent from H_1 destined to H_4 . (3 points)
{ 1 point per correct host }
Solution: H2, H3, H4
- c. If A, B, C and D are all learning bridges that know the locations of all the hosts in the network, which hosts can hear the packet sent from H_1 to H_4 ? (2 points)
{ Full grade if correct answer and 0 else }
Solution: H4

3. Consider 4 hosts A, B, C and D on a single LAN with the following ARP tables:

Host A (IP = 10.0.0.1; MAC = 0:0:1)

ARP table:

IP	MAC	Time
10.0.0.3	0:0:3	5

Host B (IP = 10.0.0.2; MAC = 0:0:2)

ARP table:

IP	MAC	Time
10.0.0.4	0:0:4	7

Host C (IP = 10.0.0.3; MAC = 0:0:3)

ARP table:

IP	MAC	Time
10.0.0.1	0:0:1	5

Host D (IP = 10.0.0.4; MAC = 0:0:4)

ARP table:

IP	MAC	Time
10.0.0.2	0:0:2	7

The third column in the ARP tables (Time) represents the number of minutes since the ARP cache entry has been updated. Suppose that one minute later, Host A performs an ARP request for Host D. When the request is completed, the ARP table of Host A becomes:

Host A (IP = 10.0.0.1; MAC = 0:0:1)

ARP table:

IP	MAC	Time
10.0.0.3	0:0:3	6
10.0.0.4	0:0:4	0

Write the contents of the ARP tables for the other three hosts B, C and D when the request is completed. (6 points)

{ 2 points for each ARP table: subtract 0.5 point if the time is wrong and 0.5 point if the host is wrong }

Host B (IP = 10.0.0.2; MAC = 0:0:2)

ARP table:

IP	MAC	Time
10.0.0.1	0:0:1	0
10.0.0.4	0:0:4	8

Host C (IP = 10.0.0.3; MAC = 0:0:3)

ARP table:

IP	MAC	Time
10.0.0.1	0:0:1	0

Host D (IP = 10.0.0.4; MAC = 0:0:4)

ARP table:

IP	MAC	Time
10.0.0.1	0:0:1	0
10.0.0.2	0:0:2	8